

**Department of Computer Science****Open Ended Lab****Course Code: CS-251L****Name: \_\_\_\_\_****Course Title: Information security Lab****Instructor name: Hassan shafqaat****Total Marks: 50****Discipline/Semester: BS(CS)- 3<sup>rd</sup> B****Date: 27-1-2026****Timing: 1.5 hours**

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**Information Security – Open Ended Coding Lab****Instructions:**

- Students must submit source code along with sample input and output.
- Explanations without working code will receive zero marks.
- Generic AI-generated solutions without customization will be penalized.

**Question 1:**

A corporate office building uses an automated access control system based on RFID cards during office hours and multi-factor authentication after office hours. The security team suspects misuse of access privileges.

**Input Format:**

timestamp, employee\_id, access\_point\_id, access\_result

**Tasks:**

1. Parse and normalize access logs.
2. Detect logically impossible access patterns.
3. Identify abnormal access frequency.
4. Compute a risk score per employee.
5. Output a ranked list of suspicious employees.

**Question 2:**

An organization allows employees to access internal resources using personal mobile devices. Despite VPN enforcement, data leakage incidents have been reported.

**Tasks:**

1. Detect dangerous permission combinations.
2. Identify abnormal background data transmission.
3. Detect potential VPN bypass behavior.
4. Generate a device-level security assessment report.

### **Question 3: Integrated Physical and Cyber Security Timeline Reconstruction**

A secure facility experienced unauthorized configuration changes without triggering alarms.

**Tasks:**

- 1. Synchronize multiple log sources.**
- 2. Detect periods of weakened security.**
- 3. Correlate physical presence with system activity.**
- 4. Identify the most probable breach window.**
- 5. Output a unified suspicious timeline.**